

Cryptography

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1 Divisibility

Def 1.1. Let $a, b \in \mathbb{Z}$ with $b \neq 0$. We say that b divides a , denoted by $b \mid a$, if there exists an integer $c \in \mathbb{Z}$ such that $a = b \cdot c$.

Prop 1.1: Transitivity of Divisibility. Let $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $b \mid c$, $\rightarrow a \mid c$.

Proof. Assume $a \mid b$. Since $a, b \in \mathbb{Z}$ and $a \neq 0$, $\exists c_1 \in \mathbb{Z} : b = a \cdot c_1$. Assume $b \mid c$. Since $b, c \in \mathbb{Z}$ and $b \neq 0$, $\exists c_2 \in \mathbb{Z} : c = b \cdot c_2$.

Substituting the first equation into the second yields:

$$c = (a \cdot c_1) \cdot c_2 = a \cdot (c_1 c_2)$$

Since $c_1, c_2 \in \mathbb{Z}$, their product $c_1 c_2$ is also an integer. Let $k = c_1 c_2 \in \mathbb{Z}$. Thus, $c = a \cdot k$. By the definition of divisibility, we conclude that $a \mid c$. $\square$

Prop 1.2: Equality from Mutual Divisibility. Let $a, b \in \mathbb{Z}$, with $a \neq 0$ and $b \neq 0$. If $a \mid b$ and $b \mid a$, $\rightarrow a = \pm b$.

Proof. Assume $a \mid b$. Since $a \neq 0$, $\exists c_1 \in \mathbb{Z} : b = a \cdot c_1$. Assume $b \mid a$. Since $b \neq 0$, $\exists c_2 \in \mathbb{Z} : a = b \cdot c_2$.

Substitute the expression for b from the first equation into the second:

$$a = (a \cdot c_1) \cdot c_2 = a \cdot (c_1 c_2)$$

Since $a \neq 0$, we can divide both sides by a , which gives:

$$1 = c_1 c_2$$

Since c_1 and c_2 are integers, the only possible pairs for (c_1, c_2) are $(1, 1)$ or $(-1, -1)$.

Case 1: $c_1 = 1$. Substituting into $b = a \cdot c_1$ gives $b = a \cdot 1 = a$. Case 2: $c_1 = -1$. Substituting into $b = a \cdot c_1$ gives $b = a \cdot (-1) = -a$.

Therefore, we conclude that $a = \pm b$. $\square$

Prop 1.3: Linearity Property of Divisibility. Let $a, b, c \in \mathbb{Z}$, with $a \neq 0$. If $a \mid b$ and $a \mid c$, $\rightarrow a \mid (b + c)$ and $a \mid (b - c)$.

Proof. Assume $a \mid b$. Since $a \neq 0$, $\exists k_1 \in \mathbb{Z} : b = a \cdot k_1$. Assume $a \mid c$. Since $a \neq 0$, $\exists k_2 \in \mathbb{Z} : c = a \cdot k_2$.

Demonstration for the Sum ($a \mid (b + c)$): Summing the two equations:

$$b + c = a \cdot k_1 + a \cdot k_2$$

Factoring out a :

$$b + c = a \cdot (k_1 + k_2)$$

Since $k_1, k_2 \in \mathbb{Z}$, their sum $(k_1 + k_2)$ is also an integer. Let $k_s = k_1 + k_2 \in \mathbb{Z}$. Thus, $b + c = a \cdot k_s$. By the definition of divisibility, we have $a \mid (b + c)$.

Demonstration for the Difference ($a \mid (b - c)$): Subtracting the second equation from the first:

$$b - c = a \cdot k_1 - a \cdot k_2$$

Factoring out a :

$$b - c = a \cdot (k_1 - k_2)$$

Since $k_1, k_2 \in \mathbb{Z}$, their difference $(k_1 - k_2)$ is also an integer. Let $k_d = k_1 - k_2 \in \mathbb{Z}$. Thus, $b - c = a \cdot k_d$. By the definition of divisibility, we have $a \mid (b - c)$. $\square$

Prop 1.4: Euclidean Division. *Let $a \in \mathbb{N}$ and $b \in \mathbb{N}$ with $b > 0$. Then $\exists! q, r \in \mathbb{Z} : a = bq + r$ with $0 \leq r < b$.*

Prop 1.5: Binary Representation from Repeated Division. *Let $m \in \mathbb{N}$, with $m < 2^B$. The repeated division-by-2 process:*

$$m = 2q_0 + m_0$$

$$q_0 = 2q_1 + m_1$$

$$q_1 = 2q_2 + m_2$$

$\dots$

terminates with $q_{B-1} = 0$, and $m = m_0 + 2m_1 + \dots + 2^{B-1}m_{B-1}$.

Proof. The proof consists of two parts: demonstrating the polynomial form and proving that the process terminates.

Polynomial Form

We can show the relationship by algebraic substitution.

$$m = 2q_0 + m_0$$

Substitute $q_0 = 2q_1 + m_1$:

$$m = 2(2q_1 + m_1) + m_0 = 2^2q_1 + 2m_1 + m_0$$

Substitute $q_1 = 2q_2 + m_2$:

$$m = 2^2(2q_2 + m_2) + 2m_1 + m_0 = 2^3q_2 + 2^2m_2 + 2m_1 + m_0$$

Continuing this process until the final step where $q_{B-2} = 2q_{B-1} + m_{B-1}$, we arrive at the final form: $m = m_0 + 2m_1 + 2^2m_2 + \dots + 2^{B-1}m_{B-1}$.

Termination

The process is guaranteed to terminate because the sequence of quotients is strictly decreasing and bounded.

- Since $m < 2^B$, the first division $m = 2q_0 + m_0$ implies $2q_0 \leq m < 2^B$, so $q_0 < 2^{B-1}$.
- Similarly, $q_0 < 2^{B-1}$ implies $2q_1 \leq q_0 < 2^{B-1}$, so $q_1 < 2^{B-2}$.

This pattern continues: $q_i < 2^{B-1-i}$. For the final quotient q_{B-1} , we have $q_{B-1} < 2^{B-1-(B-1)} = 2^0 = 1$. Since $q_{B-1} \in \mathbb{N} \cup \{0\}$, the only possibility is $q_{B-1} = 0$. The process terminates when the quotient becomes zero. $\square$

Rem: Upper Bound of a B-bit Number. *If m has B bits in its binary representation, i.e., $m = (m_{B-1} \dots m_0)_2$, then $m < 2^B$.*

Proof. The largest possible value for a B -bit number occurs when all bits are 1. This value can be calculated by summing the geometric series:

$$m \leq \sum_{i=0}^{B-1} 2^i = \frac{2^B - 1}{2 - 1} = 2^B - 1$$

Therefore, $m < 2^B$. $\square$

Cor 1.1: Range of a B-bit Number. *A natural number $m \in \mathbb{N}$ has a binary representation with B bits (where the most significant bit $m_{B-1} = 1$) $\iff 2^{B-1} \leq m < 2^B$.*

2 The Greatest Common Divisor (GCD)

Def 2.1: Greatest Common Divisor. Let $a, b \in \mathbb{Z}$, such that $a \neq 0$ or $b \neq 0$. The set of their common divisors has a largest element $d \in \mathbb{N}$, called the greatest common divisor of a and b . We write $d = \gcd(a, b)$.

Rem: Example of GCD. To find $\gcd(18, 12)$, we can list the divisors of each number:

- Divisors of 12: $\{1, 2, 3, 4, 6, 12\}$
- Divisors of 18: $\{1, 2, 3, 6, 9, 18\}$

The set of common divisors is $\{1, 2, 3, 6\}$. The largest element in this set is 6, so $\gcd(18, 12) = 6$.

While listing all divisors is feasible for small numbers, it is impractical for large integers. A far more powerful and efficient method is the Euclidean Algorithm.

3 The Euclidean Algorithm

Th 3.1: Euclidean Algorithm. *Let $a, b \in \mathbb{N} : a > 0, b > 0$ and $a \geq b$. The following algorithm computes $\gcd(a, b)$ in a finite number of steps:*

1. Let $r_0 := a$ and $r_1 := b$.
2. Set $i = 1$.
3. Divide r_{i-1} by r_i to obtain a quotient q_i and remainder r_{i+1} : $r_{i-1} = r_i q_i + r_{i+1}$.
4. If $r_{i+1} = 0$, then $r_i = \gcd(a, b)$. The algorithm terminates.
5. Otherwise, increment $i \rightarrow i + 1$ and return to Step 3.

Proof. Let the sequence of remainders be $r_0, r_1, \dots, r_n, r_{n+1}$ where $d = r_n$ is the last non-zero remainder and $r_{n+1} = 0$. We must demonstrate two facts: that d is a common divisor of a and b , and that it is the greatest common divisor.

a. $d \mid a$ and $d \mid b$

We proceed by working backward from the end of the algorithm.

- The final division step is $r_{n-1} = r_n q_n + r_{n+1}$. Since $r_{n+1} = 0$ and $d = r_n$, this equation becomes $r_{n-1} = d q_n$. This demonstrates $d \mid r_{n-1}$.
- The preceding step is $r_{n-2} = r_{n-1} q_{n-1} + r_n$. Since $d \mid r_n$ and we have shown $d \mid r_{n-1}$, it follows that d must divide their linear combination, $d \mid r_{n-2}$.

Continuing this backward induction process, we establish that d divides every preceding remainder. This concludes that $d \mid r_1$ (where $r_1 = b$) and $d \mid r_0$ (where $r_0 = a$). Thus, d is a common divisor.

b. d is the Greatest Common Divisor

Let $c \in \mathbb{Z}$ be any common divisor of a and b . We work forward down the chain of remainders.

- Since $c \mid a$ (r_0) and $c \mid b$ (r_1), and $r_2 = r_0 - r_1 q_1$, it must be that $c \mid r_2$.
- Since $c \mid r_1$ and $c \mid r_2$, and $r_3 = r_1 - r_2 q_2$, it must be that $c \mid r_3$.

Continuing this process, we eventually find that c must divide the last non-zero remainder, r_n , so $c \mid d$. Since $c \mid d$, it is a consequence of divisibility that $c \leq d$. This holds $\forall c$ common divisor, proving that d is the greatest common divisor. $\square$

Prop 3.1: Extended Euclidean Algorithm. *Let $a, b \in \mathbb{N}$ such that $a \neq 0$ or $b \neq 0$. If $d = \gcd(a, b)$, then $\exists u, v \in \mathbb{Z} : d = au + bv$.*

Proof. The proof relies on the standard Euclidean Algorithm applied to a and b , which generates

a sequence of remainders r_i :

$$\begin{aligned}
 a &= q_1b + r_1, & \text{with } 0 \leq r_1 < b \\
 b &= q_2r_1 + r_2, & \text{with } 0 \leq r_2 < r_1 \\
 r_1 &= q_3r_2 + r_3, & \text{with } 0 \leq r_3 < r_2 \\
 &\vdots \\
 r_{n-3} &= q_{n-1}r_{n-2} + r_{n-1}, & \text{with } 0 \leq r_{n-1} < r_{n-2} \\
 r_{n-2} &= q_nr_{n-1} + 0.
 \end{aligned}$$

Since r_{n-1} is the last non-zero remainder, we have $d = \gcd(a, b) = r_{n-1}$.

We proceed by ****backward substitution**** (as indicated by the notes on the right):

- From the second-to-last equation, r_{n-1} can be expressed as a linear combination of r_{n-3} and r_{n-2} :

$$d = r_{n-1} = r_{n-3} - q_{n-1}r_{n-2}$$

- Since r_{n-2} is a linear combination of r_{n-4} and r_{n-3} , substituting this expression for r_{n-2} into the equation for d will express d as a linear combination of r_{n-4} and r_{n-3} .
- By continuing this backward substitution process (i.e., replacing each remainder r_i with its expression from the previous division step in the Euclidean algorithm), we progressively express d as a linear combination of the previous two terms in the remainder sequence:

$$r_i = \text{lin. comb. of } r_{i-2} \text{ and } r_{i-1}$$

- This process continues until we reach the first two equations, where d will be expressed as a linear combination of a and b :

$$d = r_{n-1} = au + bv$$

for some integers $u, v \in \mathbb{Z}$.

□

Prop 3.2: Solvability of Linear Diophantine Equations. *Let $a, b \in \mathbb{Z}$ be integers, not both zero. The equation $au + bv = c$ has integer solutions $u, v \in \mathbb{Z}$ if and only if $\gcd(a, b) \mid c$.*

Proof. ($\Leftarrow$) **If $\gcd(a, b) \mid c$, solutions exist**

Assume $c = k \cdot \gcd(a, b)$ for some $k \in \mathbb{Z}$. By Bézout's Identity, $\exists u_0, v_0 \in \mathbb{Z}$ so that $au_0 + bv_0 = \gcd(a, b)$.

Multiplying by k yields:

$$k(au_0 + bv_0) = k \cdot \gcd(a, b) \Rightarrow a(ku_0) + b(kv_0) = c$$

Thus, $u = ku_0$ and $v = kv_0$ are integer solutions.

($\Rightarrow$) **If solutions exist, then $\gcd(a, b) \mid c$**

Assume $au + bv = c$ has integer solutions $u, v \in \mathbb{Z}$.

Let $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, d must divide any linear combination of a and b :

$$d \mid (au + bv) \Rightarrow d \mid c$$

□

Prop 3.3: All Integer Solutions to $au + bv = c$. *Let $a, b \in \mathbb{Z}$ be relatively prime (i.e., $\gcd(a, b) = 1$), and let $u_0, v_0 \in \mathbb{Z}$ be a particular solution so that $au_0 + bv_0 = c$. Then, all integer solutions (u, v) to $au + bv = c$ are of the form:*

$$u = u_0 + kb \quad \text{and} \quad v = v_0 - ka$$

for any $k \in \mathbb{Z}$.

Proof. **Part 1: Verification (The form gives solutions)**

Substituting the general form into the equation:

$$\begin{aligned} a(u_0 + kb) + b(v_0 - ka) &= au_0 + akb + bv_0 - bka \\ &= (au_0 + bv_0) + (akb - bka) \\ &= c + 0 \\ &= c \end{aligned}$$

Thus, $\forall k \in \mathbb{Z}$, the formula yields a solution.

Part 2: Completeness (All solutions must be of this form)

Assume (u, v) is any arbitrary solution, so $au + bv = c$. We also have the particular solution $au_0 + bv_0 = c$.

Subtracting the two equations gives:

$$a(u - u_0) + b(v - v_0) = 0 \Rightarrow a(u - u_0) = -b(v - v_0) \Rightarrow a(u - u_0) = b(v_0 - v)$$

Since $a \mid b(v_0 - v)$ and we are given $\gcd(a, b) = 1$, by Euclid's Lemma, it must be that a divides the other factor, $(v_0 - v)$:

$$a \mid (v_0 - v)$$

Let $k \in \mathbb{Z}$ be the integer so that $v_0 - v = ka$. Rearranging this gives the form for v :

$$v = v_0 - ka$$

Substituting $v_0 - v = ka$ back into the equality $a(u - u_0) = b(v_0 - v)$:

$$a(u - u_0) = b(ka) = a(kb)$$

Since $a \neq 0$, we can divide by a to get:

$$u - u_0 = kb$$

Rearranging this gives the form for u :

$$u = u_0 + kb$$

Hence, all solutions are of the specified form. □

4 Coprime numbers

Def 4.1: Prime Number. A natural number $p \in \mathbb{N}$ such that $p \geq 2$ is called **prime** if $\forall$ divisors $d \in \mathbb{N}$ of p , we have $d = 1$ or $d = p$.

Th 4.1: Euclid. There are infinitely many prime numbers.

Proof. We proceed by contradiction.

Assume the contrary: that the set of all prime numbers is finite.

Let $N \in \mathbb{N}$ be the largest prime number so that $\forall$ prime number p , we have $p \leq N$.

Consider the natural number $M = N! + 1$.

Since $N \geq 2$, we have $M = N! + 1 > N$. Since M is greater than the largest assumed prime N , it follows that M **is not prime**.

Therefore, M must have a prime divisor. Let $p \in \mathbb{N}$ be any prime divisor of M .

Since p is a prime number and we assumed N is the largest prime, it must be that $\forall$ prime divisor p of M , we have $p \leq N$.

Since $p \leq N$, it follows that p is one of the factors in the factorial $N! = 1 \cdot 2 \cdot 3 \cdots N$.

$$\Rightarrow p \mid N!$$

Since we defined p as a divisor of M , we also have:

$$\Rightarrow p \mid (N! + 1)$$

Since p divides both $N! + 1$ and $N!$, p must divide their difference:

$$\begin{aligned} p \mid ((N! + 1) - N!) \\ p \mid 1 \end{aligned}$$

This is a **contradiction**, because a prime number $p \geq 2$ cannot divide 1.

Therefore, the initial assumption that there exists a largest prime N must be false.

Thus, there must exist a prime number $p > N$. Since N was assumed to be the largest prime, the existence of a larger prime p proves that the set of prime numbers is **infinite**. $\square$

Prop 4.1: Divisors of Composite Numbers. If a natural number $n \in \mathbb{N}$ is not prime (i.e., composite), then n has a divisor $a \in \mathbb{N}$ such that $1 < a \leq \sqrt{n}$.

Proof. If n is not prime (composite), then n must be the product of two integer factors a and b , both greater than 1. So, there exist $a, b \in \mathbb{N}$ such that $a > 1$, $b > 1$, and $n = ab$.

We proceed by contradiction on the size of the factors. Assume the contrary: that $\forall$ factors a and b of n , we have both $a > \sqrt{n}$ and $b > \sqrt{n}$.

If $a > \sqrt{n}$ and $b > \sqrt{n}$, then multiplying these two inequalities yields:

$$\begin{aligned} n = ab &> (\sqrt{n}) \cdot (\sqrt{n}) \\ n &> (\sqrt{n})^2 \\ n &> n \end{aligned}$$

This is a **contradiction**, as n cannot be strictly greater than itself.

Therefore, the assumption that both factors a and b are greater than $\sqrt{n}$ must be false. This means that at least one of the factors must be less than or equal to $\sqrt{n}$.

Let a be the smaller of the two factors. We have $1 < a$ (by definition of a composite number) and $a \leq \sqrt{n}$. Thus, n has a divisor a such that $1 < a \leq \sqrt{n}$. $\square$

Prop 4.2: Characterisation of Coprime Numbers. *Integers $a, b \in \mathbb{Z}$ are relatively prime $\iff \exists u, v \in \mathbb{Z} : au + bv = 1$.*

Proof.

- $(\Rightarrow)$ This is a direct consequence of the Extended Euclidean Algorithm. If $\gcd(a, b) = 1$, then the algorithm guarantees $\exists u, v \in \mathbb{Z} : au + bv = 1$.
- $(\Leftarrow)$ Conversely, assume $\exists u, v \in \mathbb{Z} : au + bv = 1$. Let $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, it must divide their linear combination $au + bv$. Therefore, $d \mid 1$. The only $d \in \mathbb{N}$ so that $d \mid 1$ is $d = 1$. $\square$

Cor 4.1. *Let p be a prime number and $m \in \mathbb{Z}$. If $p \nmid m$, then $\gcd(p, m) = 1$.*

Proof. The positive divisors of a prime number p are only $\{1, p\}$. If $p \nmid m$, then p is not a common divisor. Therefore, the only common divisor of p and m is 1, which means $\gcd(p, m) = 1$. $\square$

Prop 4.3: Prime Divisor Property. *Let p be a prime and $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. Assume $p \mid ab$. If $p \mid a$, the proposition holds. If $p \nmid a$, then by the preceding corollary, $\gcd(p, a) = 1$. By the characterization of coprime numbers, $\exists u, v \in \mathbb{Z} : pu + av = 1$. Multiplying this equation by b gives $pub + abv = b$.

Since $p \mid pub$ and we are given $p \mid ab$, which implies $p \mid abv$. Since p divides both terms on the left side, it must divide their sum. Therefore, $p \mid (pub + abv) \rightarrow p \mid b$. $\square$

Th 4.2: Fundamental Theorem of Arithmetic. *Let $a \in \mathbb{N}$ so that $a \geq 2$. Then a can be factored as a product of primes, $a = p_1^{e_1} p_2^{e_2} \cdots p_m^{e_m}$, where p_i are distinct primes, and $\forall i \in \{1, \dots, m\}, e_i \geq 1$. The factorization into primes is unique modulo rearrangements of the order of the factors.*

Rem. *The exponent $e_i = \text{mult}_{p_i}(a)$ is the multiplicity of p_i in a . A generalized form of the prime factorization is given by:*

$$a = \prod_{p \text{ prime}} p^{\text{mult}_p(a)}$$

where $\text{mult}_p(a) := 0$ if p is not a prime factor of a .

5 Classes Modulo m

Prop 5.1. $\forall a \in \mathbb{Z}$ (the dividend) and $\forall m \in \mathbb{N}$ so that $m > 1$ (the modulus), $\exists! q, r \in \mathbb{Z} : a = mq + r$, with $0 \leq r < m$.

Th 5.1: Existence of a Unique Remainder. $\forall a \in \mathbb{Z}$ and $\forall m \in \mathbb{N} : m > 1$, $\exists! r \in \{0, 1, \dots, m-1\} : a \equiv r \pmod{m}$.

Proof. By Proposition 5.1 (Euclidean Division), we have $\exists! q, r \in \mathbb{Z} : a = mq + r$, where $0 \leq r < m$. Rearranging this equation gives $a - r = mq$. By the formal definition of divisibility, $m \mid (a - r)$. By the formal definition of congruence, this directly implies $a \equiv r \pmod{m}$. The uniqueness of r is guaranteed by the Euclidean Division Theorem. $\square$

Th 5.2: Congruence is an Equivalence Relation. *The relation of congruence modulo m (denoted by $\equiv \pmod{m}$) is an equivalence relation on the set of integers $\mathbb{Z}$.*

Proof. We must demonstrate that congruence modulo m satisfies reflexivity, symmetry, and transitivity.

- **Reflexivity:** $\forall a \in \mathbb{Z}, a - a = 0$. Since $m \mid 0$, $a \equiv a \pmod{m}$. The relation is reflexive.
- **Symmetry:** Assume $a \equiv b \pmod{m}$. $\Rightarrow m \mid (a - b)$, so $\exists k \in \mathbb{Z} : a - b = km$. Multiplying by -1 yields $b - a = (-k)m$. Since $-k \in \mathbb{Z}$, $\Rightarrow m \mid (b - a)$, so $b \equiv a \pmod{m}$. The relation is symmetric.
- **Transitivity:** Assume $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. $\Rightarrow \exists k_1, k_2 \in \mathbb{Z} : a - b = k_1m$ and $b - c = k_2m$. Adding the equations:

$$(a - b) + (b - c) = k_1m + k_2m$$

$$a - c = (k_1 + k_2)m$$

Since $k_1 + k_2 \in \mathbb{Z}$, $\Rightarrow m \mid (a - c)$, so $a \equiv c \pmod{m}$. The relation is transitive. $\square$

Prop 5.2: Equivalence Classes Form a Partition. *Let $\sim$ be an equivalence relation on a set X . The set of equivalence classes, $X / \sim$, forms a partition of X .*

6 Invertibility Modulo m

Prop 6.1: Compatibility. Let $a_1, a_2, b_1, b_2 \in \mathbb{Z}$. If $a_1 \equiv a_2 \pmod{m}$ and $b_1 \equiv b_2 \pmod{m}$, then:

1. $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$
2. $a_1 b_1 \equiv a_2 b_2 \pmod{m}$

Proof.

- **Addition:** Given $a_2 \equiv a_1 \pmod{m}$ and $b_2 \equiv b_1 \pmod{m}$, $\Rightarrow \exists k_1, k_2 \in \mathbb{Z} : a_2 - a_1 = k_1 m$ and $b_2 - b_1 = k_2 m$. Adding the equations yields $(a_2 - a_1) + (b_2 - b_1) = (k_1 + k_2)m$. Rearranging the terms: $(a_2 + b_2) - (a_1 + b_1) = (k_1 + k_2)m$. Since $k_1 + k_2 \in \mathbb{Z}$, this shows $m \mid ((a_2 + b_2) - (a_1 + b_1))$, so $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$.
- **Multiplication:** We express $a_2 = a_1 + k_1 m$ and $b_2 = b_1 + k_2 m$. Multiplying them gives:

$$a_2 b_2 = (a_1 + k_1 m)(b_1 + k_2 m) = a_1 b_1 + a_1 k_2 m + b_1 k_1 m + k_1 k_2 m^2$$

Factoring out m : $a_2 b_2 - a_1 b_1 = m(a_1 k_2 + b_1 k_1 + k_1 k_2 m)$. Let $K = a_1 k_2 + b_1 k_1 + k_1 k_2 m$. Since $K \in \mathbb{Z}$, $m \mid (a_2 b_2 - a_1 b_1)$, so $a_1 b_1 \equiv a_2 b_2 \pmod{m}$. $\square$

Prop 6.2. The set of integers modulo m under addition, $(\mathbb{Z}/m\mathbb{Z}, +)$, forms a **commutative group**.

Prop 6.3. When considering both addition and multiplication, $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$ forms a **commutative ring**.

Th 6.1. If an inverse exists, it is unique.

Proof. Assume an element $[a]$ has two inverses, $[a']$ and $[a'']$. By definition, $[a] \cdot [a'] = [1]$ and $[a] \cdot [a''] = [1]$. Since multiplication is associative and $[1]$ is the identity element, we can write:

$$[a'] = [a'] \cdot [1] = [a'] \cdot ([a] \cdot [a''])$$

$$[a'] = ([a'] \cdot [a]) \cdot [a''] = [1] \cdot [a''] = [a'']$$

Therefore, $[a'] = [a'']$, $\Rightarrow$ the inverse is unique. $\square$

Th 6.2: Characterization of Invertibility. *An element $[a] \in \mathbb{Z}/m\mathbb{Z}$ is invertible $\iff \gcd(a, m) = 1$.*

Proof. An element $[a]$ is invertible $\iff \exists u \in \mathbb{Z} : au \equiv 1 \pmod{m}$. This congruence is equivalent to the linear Diophantine equation $au - 1 = -mv$ for some $v \in \mathbb{Z}$, which can be rewritten as $au + mv = 1$. By Bézout's Identity, this equation has an integer solution $\iff \gcd(a, m) \mid 1$. Since $\gcd(a, m) \in \mathbb{N}$, this condition is satisfied $\iff \gcd(a, m) = 1$. $\square$

Prop 6.4. *The set of units under multiplication, $(\mathbb{Z}/m\mathbb{Z})^*$, forms a **commutative group** (the group of units modulo m).*

Prop 6.5. *$\forall p$ prime number, the commutative ring $(\mathbb{Z}/p\mathbb{Z}, +, \cdot)$ is a **field**, commonly denoted $\mathbb{F}_p$.*

Cor 6.1. *Let $p \in \mathbb{N}$ be a prime so that $p > 2$. The equation $x^2 \equiv 1 \pmod{p}$ in $\mathbb{F}_p$ has exactly two distinct solutions: $x \equiv 1 \pmod{p}$ and $x \equiv p - 1 \pmod{p}$ (i.e., $x \equiv -1 \pmod{p}$).*

Proof. The equation $x^2 \equiv 1 \pmod{p}$ is equivalent to $p \mid (x^2 - 1)$, or $p \mid (x - 1)(x + 1)$. Since p is prime, by the Prime Divisor Property:

- Either $p \mid (x - 1)$, which implies $x \equiv 1 \pmod{p}$.
- Or $p \mid (x + 1)$, which implies $x \equiv -1 \equiv p - 1 \pmod{p}$.

These two solutions are distinct because $1 \not\equiv -1 \pmod{p}$ for $p > 2$. $\square$

7 Solving Linear Congruences

Prop 7.1: Solvability Condition. *The linear congruence $ax \equiv b \pmod{m}$ has at least one integer solution for $x \in \mathbb{Z} \iff d = \gcd(a, m) \mid b$.*

Proof. The congruence $ax \equiv b \pmod{m}$ is equivalent to the linear Diophantine equation $ax + mv = b$ for some $v \in \mathbb{Z}$. This equation is solvable for $x, v \in \mathbb{Z} \iff$ the greatest common divisor of the coefficients, $d = \gcd(a, m)$, divides the constant term b . This follows directly from the Solvability Condition for Diophantine Equations (Bézout's identity). $\square$

Prop 7.2: Number of Solutions. *If $d = \gcd(a, m) \mid b$, the congruence $ax \equiv b \pmod{m}$ has exactly d distinct solutions modulo m .*

8 Chinese Remainder Theorem (CRT)

Chinese Remainder Theorem (CRT)

Th 8.1: Chinese Remainder Theorem. Let $m_1, m_2, \dots, m_k \in \mathbb{N}$ be pairwise relatively prime integers (i.e., $\forall i \neq j, \gcd(m_i, m_j) = 1$). Let $a_1, a_2, \dots, a_k \in \mathbb{Z}$ be arbitrary integers. The system of congruences:

$$x \equiv a_i \pmod{m_i} \quad \forall i \in \{1, \dots, k\}$$

has a solution, and this solution is unique modulo $M = \prod_{i=1}^k m_i$.

Proof. The proof is constructive for existence and uses coprimality for uniqueness. Let $M = \prod_{i=1}^k m_i$ and $N_j = M/m_j$.

Existence (Construction)

Since $\gcd(N_j, m_j) = 1$, $\exists v_j \in \mathbb{Z}$ so that $v_j N_j \equiv 1 \pmod{m_j}$. We construct the solution x :

$$x = \sum_{j=1}^k a_j v_j N_j$$

For an arbitrary index i , consider $x \pmod{m_i}$. $\forall j \neq i, m_i \mid N_j \implies N_j \equiv 0 \pmod{m_i}$. Thus, all terms $j \neq i$ vanish modulo m_i :

$$x \equiv a_i v_i N_i + \sum_{j \neq i} (0) \equiv a_i (1) \equiv a_i \pmod{m_i}$$

A solution x exists.

Uniqueness (Modulo M)

Assume x and x' are two solutions. Since $x \equiv a_i \pmod{m_i}$ and $x' \equiv a_i \pmod{m_i}$ for all i , we have $x - x' \equiv 0 \pmod{m_i}$, meaning $m_i \mid (x - x') \forall i$. Since the m_i are pairwise coprime, their product $M = \prod m_i$ must also divide the difference:

$$M \mid (x - x') \implies x \equiv x' \pmod{M}$$

The solution is unique modulo M . □

9 Euler's ϕ Function

Def 9.1: Euler's Totient Function, ϕ . Let $m \in \mathbb{N}$ such that $m \geq 1$. The **Euler's totient function**, denoted by $\phi(m)$, is the number of elements in the multiplicative group of integers modulo m , which is $(\mathbb{Z}/m\mathbb{Z})^*$.

In other words, $\phi(m)$ is the number of integers k such that $1 \leq k \leq m$ so that $\gcd(k, m) = 1$.

Def 9.2: Remainder Map. Let $m, n \in \mathbb{N}$ be integers. The **remainder map** r is a function:

$$r : \mathbb{Z}/(mn)\mathbb{Z} \longrightarrow (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$$

It is defined $\forall a \in \{0, \dots, mn - 1\}$ as:

$$r([a]_{mn}) := ([a]_m, [a]_n)$$

where $[a]_k$ denotes the residue class of a modulo k .

Prop 9.1: Remainder Map is Well-Defined. The remainder map r is well-defined. That is, the result of the map is independent of the choice of the representative of the residue class:

$$[a]_{mn} = [b]_{mn} \implies r([a]_{mn}) = r([b]_{mn})$$

which means $([a]_m, [a]_n) = ([b]_m, [b]_n)$.

Proof. We assume the antecedent: $[a]_{mn} = [b]_{mn}$.

By the definition of equality of residue classes, this means mn divides the difference $(b - a)$:

$$[a]_{mn} = [b]_{mn} \implies mn \mid (b - a)$$

Since $mn \mid (b - a)$, it must be that both m and n individually divide the difference $(b - a)$:

$$mn \mid (b - a) \implies \begin{cases} m \mid (b - a) \\ n \mid (b - a) \end{cases}$$

By the definition of congruence and residue classes, this translates back into the desired equalities:

$$\begin{cases} m \mid (b - a) \implies [a]_m = [b]_m \\ n \mid (b - a) \implies [a]_n = [b]_n \end{cases}$$

Since both components are equal, the ordered pairs are equal:

$$([a]_m, [a]_n) = ([b]_m, [b]_n)$$

Thus, $r([a]_{mn}) = r([b]_{mn})$, and the map is well-defined. □

Th 9.1: Bijectivity of the Remainder Map with Coprime Factors. *Let $m, n \in \mathbb{N}$ be integers such that $m, n \geq 1$, and assume m and n are **pairwise coprime** (i.e., $\gcd(m, n) = 1$).*

The remainder map r :

$$r : \mathbb{Z}/(mn)\mathbb{Z} \longrightarrow (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$$

is a **bijection** (one-to-one and onto).

Proof. The proof establishes bijectivity by demonstrating both **Surjectivity** (the map is onto) and **Injectivity** (the map is one-to-one).

Surjectivity (Onto)

Let $([a]_m, [b]_n) \in (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$ be an arbitrary element in the codomain.

We seek to show $\exists x \in \mathbb{Z}$ so that $r([x]_{mn}) = ([a]_m, [b]_n)$.

This is equivalent to finding an integer x that simultaneously solves the system of congruences:

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases}$$

Since we are given that $\gcd(m, n) = 1$, the **Chinese Remainder Theorem** guarantees that a solution x always exists for any pair of residues (a, b) .

Therefore, $\forall$ element in the codomain, there is a corresponding pre-image in the domain, proving that the map r is **surjective**.

Injectivity (One-to-One)

We can establish injectivity by counting the elements (since the sets are finite):

1. **Count the Domain:** The number of elements in the domain is the modulus:

$$|\mathbb{Z}/(mn)\mathbb{Z}| = mn$$

2. **Count the Codomain:** The number of elements in the codomain (a Cartesian product) is the product of the sizes of the factors:

$$|(\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})| = |\mathbb{Z}/m\mathbb{Z}| \cdot |\mathbb{Z}/n\mathbb{Z}| = m \cdot n = mn$$

3. **Conclusion:** Since the domain and codomain are finite sets and have the **same number of elements** ($mn = mn$), and we have already proven the map r is **surjective**, it must automatically be **injective** as well.

$$\text{Surjectivity} \wedge (|\text{Domain}| = |\text{Codomain}|) \implies \text{Injectivity}$$

Since r is both surjective and injective, it is a **bijection**. □

Th 9.2: Totient of a Prime Number. *The Totient of a Prime Number: If $m \in \mathbb{N}$ so that m is prime, then $\varphi(m) = m - 1$.*

Th 9.3: Primality Test via the Totient Function. *A Primality Test via the Totient Function: $\forall m \in \mathbb{N} : m \geq 2, \varphi(m) = m - 1 \iff m$ is a prime number.*

Proof.

- ($\Rightarrow$) (“Only if” direction): We prove the contrapositive. Assume m is composite (not prime).
 $\Rightarrow \exists p \in \mathbb{N} : 1 < p < m$ and $p \mid m$. Since $p \mid m$, $\gcd(p, m) = p > 1$. Thus, p is not coprime to m . Since $p \in \{1, \dots, m - 1\}$ is excluded from the count, the total count $\varphi(m)$ must be strictly less than $m - 1$. More precisely, $\varphi(m) \leq m - 2$. This shows that the equality $\varphi(m) = m - 1$ holds $\Rightarrow m$ is prime.
- ($\Leftarrow$) (“If” direction): Assume m is prime. Then $\forall a \in \{1, \dots, m - 1\}, \gcd(a, m) = 1$. Since there are $m - 1$ such integers, $\varphi(m) = m - 1$. $\square$

Prop 9.2: Totient of a Prime Power. $\forall p$ prime and $\forall \alpha \in \mathbb{N} : \alpha \geq 1$:

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$$

Proof. The totient function $\varphi(p^\alpha)$ counts the number of integers $a \in \{1, \dots, p^\alpha\}$ so that $\gcd(a, p^\alpha) = 1$. The integers that are **not** coprime to p^α are those a for which $p \mid a$. We count these integers (the complement). These non-coprime integers are the multiples of p in the range $[1, p^\alpha]$, i.e., of the form $k \cdot p$ where $k \in \mathbb{N}$. The condition $1 \leq k \cdot p \leq p^\alpha$ implies $1 \leq k \leq p^{\alpha-1}$. Thus, there are exactly $p^{\alpha-1}$ multiples of p in the specified range. Subtracting this count from the total number of integers, p^α , yields:

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p - 1)$$

$\square$

Th 9.4: Multiplicativity of Euler’s Totient Function. *Multiplicativity of Euler’s Totient Function: If $m, n \in \mathbb{N}$ are coprime integers ($\gcd(m, n) = 1$), then $\varphi(mn) = \varphi(m)\varphi(n)$.*

10 The fast powering algorithm

Th 10.1: Solvability of a System of Two Linear Congruences. *Let m and n be positive integers, and let a and b be any integers. The system of linear congruences:*

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases}$$

has a solution if and only if $\gcd(m, n) \mid (a - b)$.

1: Constructive Proof by Cases (Alessandra's Method). We prove the biconditional nature by demonstrating necessity ($\Rightarrow$) and sufficiency ($\Leftarrow$) separately.

1. Necessity ($\Rightarrow$): We assume a solution exists $\rightarrow \exists x_0 \in \mathbb{Z} : x_0 \equiv a \pmod{m}$ and $x_0 \equiv b \pmod{n}$. This implies:

$$\begin{aligned} x_0 - a &= mk \quad \text{for some } k \in \mathbb{Z} \\ x_0 - b &= nl \quad \text{for some } l \in \mathbb{Z} \end{aligned}$$

Thus, $x_0 = a + mk$ and $x_0 = b + nl$. Equating the two expressions for x_0 yields $a + mk = b + nl$. Rearranging the terms: $a - b = nl - mk$. Let $d = \gcd(m, n)$. By definition, $d \mid m$ and $d \mid n$. Therefore, d must divide any integer linear combination of m and n , $\forall k, l \in \mathbb{Z}$. Since $a - b = nl - mk$, we conclude that $d \mid (a - b)$.

2. Sufficiency ($\Leftarrow$): We assume $\gcd(m, n) \mid (a - b)$. Let $d = \gcd(m, n)$. Since $d \mid (a - b)$, we have $a - b = kd$ for some integer $k \in \mathbb{Z}$. By Bezout's identity, $\exists u, v \in \mathbb{Z} : d = mu + nv$. Substituting the expression for d : $a - b = k(mu + nv) \rightarrow a - b = kmu + knv$. Rearranging the equation: $a - kmu = b + knv$. Let $x_0 = a - kmu$. By construction: $x_0 = a - kmu \rightarrow x_0 - a = (-ku)m$. Since $-ku \in \mathbb{Z}$, $x_0 \equiv a \pmod{m}$. Also, $x_0 = b + knv \rightarrow x_0 - b = (kn)v$. Since $kn \in \mathbb{Z}$, $x_0 \equiv b \pmod{n}$. Since x_0 satisfies both congruences, a solution exists. $\square$

2: Direct Proof via Reduction (Professor's Method). The first congruence $x \equiv a \pmod{m}$ can be rewritten as $x = a + km$ for some integer $k \in \mathbb{Z}$. Substituting this into the second congruence yields:

$$a + km \equiv b \pmod{n}$$

Rearranging the terms, we obtain a single linear congruence for the unknown k :

$$km \equiv b - a \pmod{n}$$

A solution for this linear congruence exists if and only if $\gcd(m, n) \mid (b - a)$. Since $\gcd(m, n) = \gcd(m, n)$ and $\gcd(m, n) \mid (b - a)$ is equivalent to $\gcd(m, n) \mid (a - b)$, the original system has a solution if and only if the condition holds. $\square$

Prop 10.1: General Solvability Condition. *A system of congruences $x \equiv a_i \pmod{m_i}$ for $i \in \{1, 2, \dots, n\}$ has a solution if and only if for every pair of indices $i \neq j$, the condition $\gcd(m_i, m_j) \mid (a_j - a_i)$ holds.*

A Method to Compute $g^A \pmod{N}$

Alg 10.1: Fast Powering Algorithm. Let $g, A, N \in \mathbb{N}$ such that $g \geq 1$, $A \geq 1$, and $N \geq 1$.

Let $A = (A_r A_{r-1} \dots A_0)_2$ be the binary representation of A , so that $A = A_0 + 2A_1 + 2^2 A_2 + \dots + 2^r A_r$, where $A_i \in \{0, 1\}$ for $0 \leq i \leq r$, and $A_r = 1$.

1. **Compute the powers $g^{2^i} \pmod{N}$ for $0 \leq i \leq r$ by successive squaring:**

$$\begin{aligned} a_0 &\equiv g^{2^0} \equiv g \pmod{N} \\ a_1 &\equiv a_0^2 \equiv g^{2^1} \pmod{N} \\ a_2 &\equiv a_1^2 \equiv g^{2^2} \pmod{N} \\ &\vdots \\ a_r &\equiv a_{r-1}^2 \equiv g^{2^r} \pmod{N} \end{aligned}$$

This step requires r multiplications. The next one is the square of the previous one.

2. **Compute $g^A \pmod{N}$ using the formula $g^A = g^{A_0 + 2A_1 + 2^2 A_2 + \dots + 2^r A_r}$:**

$$\begin{aligned} g^A &\equiv g^{A_0} \cdot (g^2)^{A_1} \cdot (g^{2^2})^{A_2} \dots (g^{2^r})^{A_r} \pmod{N} \\ &\equiv a_0^{A_0} \cdot a_1^{A_1} \cdot a_2^{A_2} \dots a_r^{A_r} \pmod{N} \end{aligned}$$

This step requires at most r multiplications (since $A_i \in \{0, 1\}$).

In total (once the binary form of A is known), we need $2r \approx 2 \log_2 A$ steps instead of A (naive method)!

Number of Operations

Rem: Complexity of the Fast Powering Algorithm. If $2^r \leq A < 2^{r+1}$, then r is the number of bits in the binary representation of A ($\approx \log_2 A$).

1. It requires at most r divisions to write A in binary ($\sim r$ bits).
2. To compute g^A , where $A = A_0 + 2A_1 + 2^2 A_2 + \dots + 2^r A_r$, we need:
 - r multiplications for the successive squaring (Step 1).
 - At most r multiplications for the final product (Step 2).

Thus, we need $\sim 2r$ multiplications to compute $g^A \pmod{N}$.

Since $2^r \leq A$, it follows that $r \leq \log_2(A)$. Therefore, the total number of multiplications is $\leq 2r \leq 2 \log_2(A)$.

Prop 10.2: Efficiency Comparison. If $A \approx 2^{1000} \approx 10^{300}$, then $\log_2(A) \approx 1000$. The Fast Powering Algorithm requires $\sim 2 \log_2(A) \approx 2000$ multiplications.

In contrast, the Naive Method requires $A \approx 10^{300}$ multiplications. The Fast Powering Algorithm is dramatically more efficient.

Rem: Problem Statement. The algorithm is designed to efficiently calculate the value of $g^A \pmod{N}$, given integers $g > 1$, $A > 1$, and $N > 1$. The challenge arises when the exponent A is very large, $\forall A \in \mathbb{N} : A \gg 1$. The naive method requires $\mathcal{O}(A)$ multiplications, which is computationally infeasible for large A .

Rem: Method Principle. The Fast Powering Algorithm leverages the binary representation of the exponent A . Let the binary expansion be $A = \sum_{i=0}^r A_i 2^i$, where $A_i \in \{0, 1\}$. The exponentiation property allows us to write:

$$g^A = g^{\sum_{i=0}^r A_i 2^i} = \prod_{i=0}^r (g^{2^i})^{A_i} \pmod{N}$$

The core idea is **Successive Squaring**: the terms $g^{2^i} \pmod{N}$ are computed efficiently by starting with g^1 and repeatedly squaring the previous result, $\forall i \in \{1, \dots, r\} : g^{2^i} = (g^{2^{i-1}})^2 \pmod{N}$. The final result is the product of only those terms $g^{2^i} \pmod{N}$ for which the corresponding binary coefficient $A_i = 1$.

Rem: Complexity Analysis. The Fast Powering Algorithm significantly reduces complexity from $\mathcal{O}(A)$ (naive method) to $\mathcal{O}(\log_2(A))$. This is due to the three phases:

- **Binary Conversion:** $\sim \log_2(A)$ divisions.
- **Successive Squaring:** $\sim \log_2(A)$ squarings.
- **Final Product:** at most $\sim \log_2(A)$ multiplications.

The total number of operations $\sim 3 \log_2(A)$, providing immense efficiency gain, $\forall A \in \mathbb{N} : A \approx 2^{1000} \rightarrow \text{Operations} \approx 3000$.

Th 10.2: Fermat's Little Theorem - Primary Form. Let p be a prime number. For any integer $a \in \mathbb{Z}$ so that p does not divide a , we have:

$$a^{p-1} \equiv 1 \pmod{p}$$

Th 10.3: Fermat's Little Theorem - Alternative Form. *Let p be a prime number. For any integer $a \in \mathbb{Z}$, we have:*

$$a^p \equiv a \pmod{p}$$

Rem: Conditions for Application. *The theorem is generally false for composite moduli. The primary condition is that the modulus p must be a prime number.*

11 Fermat's little theorem

Th 11.1: Fermat's Little Theorem. *Let p be a prime. If $a \in (\mathbb{Z}/p\mathbb{Z})^*$, then $a^{p-1} = 1$. Equivalently, if p is a prime and a is an integer so that p does not divide a , then $a^{p-1} \equiv 1 \pmod{p}$.*

Proof. The proof employs a technique involving the set of invertible elements modulo p .

Step 1: Construct the Set Let $a \in (\mathbb{Z}/p\mathbb{Z})^*$. Consider the set S formed by multiplying a by each of the $p - 1$ invertible elements of $\mathbb{Z}/p\mathbb{Z}$, which are $I = \{1, 2, \dots, p - 1\}$.

$$S = \{a \cdot i \pmod{p} \mid \forall i \in I\}$$

Step 2: Prove Non-Zero Property We claim that $a \cdot i \not\equiv 0 \pmod{p}$ for all $i \in I$. Assume for contradiction that $a \cdot i \equiv 0 \pmod{p}$. This implies $p \mid (a \cdot i)$. Since p is prime, $p \mid a$ or $p \mid i$. By premise, $a \in (\mathbb{Z}/p\mathbb{Z})^*$, so $p \nmid a$. Since $1 \leq i \leq p - 1$, $\gcd(i, p) = 1$, so $p \nmid i$. This is a contradiction, so $\forall i \in I, a \cdot i \not\equiv 0 \pmod{p}$.

Step 3: Prove Distinctness We claim that all $p - 1$ elements in S are distinct. Assume for contradiction that $\exists i, j \in I$ so that $i \neq j$ and $a \cdot i \equiv a \cdot j \pmod{p}$. This implies $a \cdot (i - j) \equiv 0 \pmod{p}$. Since $p \nmid a$, we must have $p \mid (i - j)$. However, since $i, j \in \{1, \dots, p - 1\}$ and $i \neq j$, the difference $i - j$ satisfies $0 < |i - j| < p$. Such an integer cannot be a multiple of p . This contradiction proves that all elements in S are distinct.

Step 4: Equate Sets Since S contains $p - 1$ distinct, non-zero elements modulo p , and the set of all non-zero elements is $(\mathbb{Z}/p\mathbb{Z})^* = I = \{1, 2, \dots, p - 1\}$, the two sets must be identical, $\forall i \in I : S = I$.

Step 5: The Trick (Multiplication) The product of the elements in S must equal the product of the elements in I :

$$\prod_{i=1}^{p-1} (a \cdot i) \equiv \prod_{i=1}^{p-1} i \pmod{p}$$

Step 6: Simplify and Conclude Factoring the left-hand side yields:

$$a^{p-1} \cdot (p - 1)! \equiv (p - 1)! \pmod{p}$$

Since p is prime, $(p - 1)!$ is invertible modulo p (by Wilson's Theorem, or simply because $\forall i \in \{1, \dots, p - 1\}, \gcd(i, p) = 1$). We can safely cancel $(p - 1)!$ from both sides:

$$a^{p-1} \equiv 1 \pmod{p}$$

□

Cor 11.1: Modular Exponentiation. *Let p be a prime, and $a \in \mathbb{Z}$. If $\forall x, y \in \mathbb{N} \setminus \{0\}$ so that $x \equiv y \pmod{p - 1}$, then $a^x \equiv a^y \pmod{p}$.*

Proof. We consider two distinct cases for the integer a .

Case 1: p divides a If $p \mid a$, then $a \equiv 0 \pmod{p}$. Since $x \geq 1$ and $y \geq 1$, we have $a^x \equiv 0 \pmod{p}$ and $a^y \equiv 0 \pmod{p}$. Thus, $a^x \equiv a^y \pmod{p}$ holds trivially.

Case 2: p does not divide a The congruence $x \equiv y \pmod{p-1}$ means $y = x + k(p-1)$ for some $k \in \mathbb{Z}$. Substituting this into a^y :

$$a^y = a^{x+k(p-1)} = a^x \cdot a^{k(p-1)} = a^x \cdot (a^{p-1})^k$$

By Fermat's Little Theorem, $a^{p-1} \equiv 1 \pmod{p}$. Therefore:

$$a^y \equiv a^x \cdot (1)^k \pmod{p} \rightarrow a^y \equiv a^x \pmod{p}$$

□

Cor 11.2: Finding Modular Inverses. Let p be a prime and $a \in (\mathbb{Z}/p\mathbb{Z})^*$. Then the inverse of a modulo p is a^{p-2} .

Proof. We must show that $a \cdot a^{p-2} \equiv 1 \pmod{p}$.

$$a \cdot a^{p-2} = a^{1+p-2} = a^{p-1}$$

By Fermat's Little Theorem, $\forall a \in \mathbb{Z}$ so that $p \nmid a$, we have $a^{p-1} \equiv 1 \pmod{p}$. Therefore:

$$a \cdot a^{p-2} \equiv 1 \pmod{p}$$

This confirms that a^{p-2} is the multiplicative inverse of a modulo p .

□

Cor 11.3: Fermat Primality Test. Let p be an integer so that $p \geq 2$, and let a be an integer so that p does not divide a . If $a^{p-1} \not\equiv 1 \pmod{p}$, then p is not a prime number.

Proof. This statement is the contrapositive of Fermat's Little Theorem: $(\neg \text{THEN}) \Rightarrow (\neg \text{IF})$. The original theorem states: IF p is prime, THEN $a^{p-1} \equiv 1 \pmod{p}$. The contrapositive is: IF $a^{p-1} \not\equiv 1 \pmod{p}$, THEN p is not prime. This conclusion is logically sound. □

12 The Order of an Element and Primitive Roots

Def 12.1: Order of an Element. Let p be a prime and $a \in \mathbb{Z}$ so that p does not divide a . The order of a modulo p , denoted $\text{ord}_p(a)$, is the minimum positive integer $n \in \mathbb{N} \setminus \{0\}$ so that $a^n \equiv 1 \pmod{p}$.

Prop 12.1: Order Divisibility Property. Let p be a prime, $a \in \mathbb{Z}$ so that p does not divide a , and $n \geq 1$. Then $a^n \equiv 1 \pmod{p}$ if and only if $\text{ord}_p(a)$ divides n .

Proof. We demonstrate both directions of the biconditional.

First Direction ($\Leftarrow$): Assume $\text{ord}_p(a) \mid n$. $\rightarrow \exists k \in \mathbb{Z}$ so that $n = k \cdot \text{ord}_p(a)$. Then:

$$a^n = a^{k \cdot \text{ord}_p(a)} = (a^{\text{ord}_p(a)})^k$$

By definition of the order, $a^{\text{ord}_p(a)} \equiv 1 \pmod{p}$. Therefore:

$$a^n \equiv (1)^k \equiv 1 \pmod{p}$$

Second Direction ($\Rightarrow$): Assume $a^n \equiv 1 \pmod{p}$. By the division algorithm, $\exists q, r \in \mathbb{Z}$ so that $n = q \cdot \text{ord}_p(a) + r$, where $0 \leq r < \text{ord}_p(a)$. We evaluate a^n :

$$a^n = a^{q \cdot \text{ord}_p(a) + r} = (a^{\text{ord}_p(a)})^q \cdot a^r$$

Substituting the assumed and known congruences:

$$1 \equiv (1)^q \cdot a^r \pmod{p} \rightarrow a^r \equiv 1 \pmod{p}$$

Since $\text{ord}_p(a)$ is the *minimum* positive integer for which this holds, and $0 \leq r < \text{ord}_p(a)$, the only possibility is that $r = 0$. $\rightarrow \text{ord}_p(a)$ divides n . $\square$

Prop 12.2: Equality of Powers. Let p be a prime and $g \in (\mathbb{Z}/p\mathbb{Z})^*$. Then $g^x = g^y$ if and only if $x \equiv y \pmod{\text{ord}_p(g)}$.

Proof. Since $g \in (\mathbb{Z}/p\mathbb{Z})^*$, g is invertible.

$$g^x = g^y \iff g^x \cdot g^{-y} = 1 \iff g^{x-y} = 1$$

Applying the proposition from the previous section (2.1), $g^n = 1$ if and only if $\text{ord}_p(g) \mid n$. Let $n = x - y$. We conclude:

$$g^{x-y} = 1 \iff \text{ord}_p(g) \mid (x - y)$$

By the definition of modular congruence, this is equivalent to: $x \equiv y \pmod{\text{ord}_p(g)}$. $\square$

Def 12.2: Primitive Root. An element $g \in (\mathbb{Z}/p\mathbb{Z})^*$ is a primitive root if its powers generate all the elements of $(\mathbb{Z}/p\mathbb{Z})^*$. That is, the set $\{g^1, g^2, \dots, g^{p-1}\}$ is equal to the set $\{1, 2, \dots, p-1\}$.

Th 12.1: Primitive Root Condition. An element $g \in (\mathbb{Z}/p\mathbb{Z})^*$ is a primitive root if and only if $\text{ord}_p(g) = p - 1$.

Proof Outline. For g to be a primitive root, the $p - 1$ powers $\{g^1, g^2, \dots, g^{p-1}\}$ must all be distinct. By the Proposition on the Condition for Equality of Powers, $g^i = g^j$ if and only if $i \equiv j \pmod{\text{ord}_p(g)}$. For the powers to be distinct for all $1 \leq i < j \leq p - 1$, we must ensure that $g^{j-i} \neq 1$ for any exponent $j - i$ where $1 \leq j - i < p - 1$. This means that the smallest positive exponent k so that $g^k = 1$ must be $p - 1$. By definition, $\text{ord}_p(g) = k$. Therefore, g is a primitive root if and only if $\text{ord}_p(g) = p - 1$. $\square$

Def 12.3. Let p be a prime, and $a \in \mathbb{Z}$ such that $p \nmid a$. The **order** $\text{ord}_p(a)$ of a modulo p is the minimum $n \in \mathbb{N}$ such that $n \geq 1$ and $a^n \equiv 1 \pmod{p}$.

Prop 12.3. Let p be a prime, $a \in \mathbb{Z}$ such that $p \nmid a$, and $n \geq 1$. Then $a^n \equiv 1 \pmod{p}$ if and only if the order of a modulo p , denoted $\text{ord}_p(a)$, divides n . In particular, $\text{ord}_p(a) \mid (p - 1)$ (a consequence of Fermat's Little Theorem).

Proof. Let $k = \text{ord}_p(a)$.

$(\Rightarrow)$ If $a^n \equiv 1 \pmod{p}$, then $k \mid n$

Use the Division Algorithm on n by k : $n = qk + r$, where $0 \leq r < k$.

$$1 \equiv a^n \equiv a^{qk+r} \equiv (a^k)^q \cdot a^r \equiv 1^q \cdot a^r \equiv a^r \pmod{p}$$

Since $a^r \equiv 1 \pmod{p}$ and k is the **minimum positive exponent** satisfying this property, and $r < k$, we must have $r = 0$. Thus, $n = qk$, which means $k \mid n$.

$(\Leftarrow)$ If $k \mid n$, then $a^n \equiv 1 \pmod{p}$

If $k \mid n$, then $n = qk$ for some $q \in \mathbb{Z}$.

$$a^n = a^{qk} = (a^k)^q \equiv 1^q \equiv 1 \pmod{p}$$

$\square$

Cor 12.1. Let p be a prime, $a \in \mathbb{Z}$ such that $p \nmid a$. $\text{ord}_p(a) = \min\{n \in \mathbb{N} : n \geq 1 \text{ so that } a^n \equiv 1 \pmod{p}\} = \min\{n \in \mathbb{N} : n \geq 1 \text{ so that } a^n \equiv 1 \pmod{p} \text{ and } n \mid p - 1\}$.

Prop 12.4. Let p be a prime. If $g \in \mathbb{F}_p^*$, then $g^x = g^y \iff y \equiv x \pmod{\text{ord}_p(g)}$.

Proof. ($\Leftarrow$) Assume $x \equiv y \pmod{\text{ord}_p(g)}$. Then $x - y = k \cdot \text{ord}_p(g)$ for some integer k .
 $g^x = g^{y+k \cdot \text{ord}_p(g)} = g^y \cdot (g^{\text{ord}_p(g)})^k \equiv g^y \cdot 1^k \equiv g^y \pmod{p}$.

($\Rightarrow$) Assume $g^x = g^y$. Without loss of generality, assume $x \geq y$. Since $g \in \mathbb{F}_p^*$ is invertible, multiplying by $(g^{-1})^y$ gives $g^{x-y} = 1$. By the previous Proposition, $\text{ord}_p(g) \mid x - y$, which means $x \equiv y \pmod{\text{ord}_p(g)}$. $\square$

Def 12.4. Let p be a prime. A **primitive root** of $\mathbb{F}_p$ is any $g \in \mathbb{F}_p^*$ such that the powers of g generate all the elements of $\mathbb{F}_p^*$, i.e., $\mathbb{F}_p^* = \{1, g, g^2, \dots, g^{p-2}\}$.

Th 12.2: Characterization of the Primitive Roots. Let p be a prime. $g \in \mathbb{F}_p^*$ is a primitive root if and only if $\text{ord}_p(g) = p - 1$.

Proof. The proof establishes the equivalence between g generating the group $\mathbb{F}_p^*$ and g having the maximum possible order. Note that $|\mathbb{F}_p^*| = p - 1$.

($\Rightarrow$) **If g is a primitive root, then $\text{ord}_p(g) = p - 1$**

By definition, if g is a primitive root, the set of its powers $\{1, g, \dots, g^{p-2}\}$ contains **all** $p - 1$ **distinct elements** of $\mathbb{F}_p^*$.

Let $m = \text{ord}_p(g)$. If we assume $m < p - 1$, then the sequence of powers $1, g, g^2, \dots$ repeats with a period of m . This means the set of distinct powers $\langle g \rangle$ has size $|\langle g \rangle| = m$.

Since g is a primitive root, $\langle g \rangle = \mathbb{F}_p^*$, so $m = |\mathbb{F}_p^*| = p - 1$. The assumption $m < p - 1$ leads to the contradiction $|\langle g \rangle| < |\mathbb{F}_p^*|$. Thus, $m = p - 1$.

($\Leftarrow$) **If $\text{ord}_p(g) = p - 1$, then g is a primitive root**

Assume $\text{ord}_p(g) = p - 1$. We must show the powers $1, g, \dots, g^{p-2}$ are all distinct.

Suppose, for contradiction, that $g^i \equiv g^j \pmod{p}$ for $0 \leq i < j \leq p - 2$. Since g is invertible, $g^{j-i} \equiv 1 \pmod{p}$.

Let $k = j - i$. Then $0 < k < p - 1$. This implies that a power $g^k \equiv 1 \pmod{p}$ for an exponent k smaller than $p - 1$. This contradicts the assumption that $\text{ord}_p(g) = p - 1$ is the **minimum** positive exponent that results in 1.

Thus, the $p - 1$ powers $\{1, g, \dots, g^{p-2}\}$ are distinct. Since $\mathbb{F}_p^*$ has exactly $p - 1$ elements, the powers must generate the entire group, meaning g is a primitive root. $\square$

Th 12.3: Existence of Primitive Roots. Let p be a prime. $\mathbb{F}_p$ has at least one primitive root. (Admitted)

Proof. Let $x = g^{\frac{p-1}{2}}$. We first observe the square of x :

$$x^2 = \left(g^{\frac{p-1}{2}}\right)^2 = g^{p-1}$$

By **Fermat's Little Theorem**, $g^{p-1} \equiv 1 \pmod{p}$, provided $p \nmid g$ (which is true since $g \in \mathbb{F}_p^*$). Thus, x satisfies the congruence:

$$x^2 \equiv 1 \pmod{p}$$

In the field $\mathbb{F}_p$, the only solutions to $x^2 \equiv 1$ are $x = 1$ and $x = -1$ (or $x \equiv p-1 \pmod{p}$).

We must show that $x \neq 1$. We proceed by contradiction.

Assume $x = 1$, meaning:

$$g^{\frac{p-1}{2}} \equiv 1 \pmod{p}$$

By the property of the order of an element, this would imply:

$$\text{ord}_p(g) \mid \frac{p-1}{2}$$

However, since g is a **primitive root**, we know by definition that $\text{ord}_p(g) = p-1$. The divisibility condition becomes:

$$(p-1) \mid \frac{p-1}{2}$$

Since $p > 2$, this condition is false (a positive integer cannot divide a number half its size). This is a **contradiction**.

Therefore, the only remaining possibility for the value of x is $x = -1$.

$$g^{\frac{p-1}{2}} \equiv -1 \pmod{p}$$

□

Rem. Let $p > 2$ be a prime. If g is a square in $\mathbb{F}_p$, then g is not a primitive root in $\mathbb{F}_p$.

Proof. Let g be a square, so $g = h^2$ for some $h \in \mathbb{F}_p^*$. Then $g^{(p-1)/2} = (h^2)^{(p-1)/2} = h^{p-1} \equiv 1 \pmod{p}$ by Fermat's Little Theorem. Since $g^{(p-1)/2} \equiv 1 \pmod{p}$, by the proposition on order, $\text{ord}_p(g) \mid \frac{p-1}{2}$. Since $\frac{p-1}{2} < p-1$ (as $p > 2$), we have $\text{ord}_p(g) < p-1$. By the characterization theorem, g is not a primitive root. □

Lemma 12.1: Effective Test for a Primitive Root Candidate. Let $p > 2$ be prime and let $p-1 = \prod_{i=1}^m p_i^{a_i}$ be the prime factorization of $p-1$. Then $g \in \mathbb{F}_p^*$ is a primitive root if and only if $g^{(p-1)/p_i} \not\equiv 1 \pmod{p}$ for all $i = 1, \dots, m$.

Proof. Let $k = \text{ord}_p(g)$. Recall that g is a primitive root $\iff k = p - 1$.

($\Rightarrow$) If g is a primitive root, the test passes

Assume g is a primitive root, so $k = p - 1$.

Suppose, for contradiction, that $g^{(p-1)/p_i} \equiv 1 \pmod{p}$ for some prime factor p_i . By the property of the order, k must divide the exponent:

$$k \mid \frac{p-1}{p_i} \implies (p-1) \mid \frac{p-1}{p_i}$$

Since p_i is a prime ($p_i \geq 2$), this divisibility is impossible. Thus, $g^{(p-1)/p_i} \not\equiv 1 \pmod{p}$ must hold for all i .

($\Leftarrow$) If the test passes, g is a primitive root

Assume $g^{(p-1)/p_i} \not\equiv 1 \pmod{p}$ for all i . We know $k \mid p - 1$, so k is a divisor of $p - 1$, and thus k must be of the form $k = p_1^{f_1} \cdots p_m^{f_m}$ with $0 \leq f_i \leq a_i$.

Suppose, for contradiction, that $k < p - 1$. This means at least one exponent f_i must be strictly less than a_i (i.e., $f_i \leq a_i - 1$).

For such an index i , we can construct the exponent $\frac{p-1}{p_i}$. Since $p_i^{a_i}$ is a factor of $p - 1$ and $p_i^{f_i}$ is a factor of k , the difference in the exponent is:

$$\frac{p-1}{p_i} = \frac{p_1^{a_1} \cdots p_i^{a_i-1} \cdots p_m^{a_m}}{p_i}$$

Since $f_i \leq a_i - 1$, we have $k \mid \frac{p-1}{p_i}$.

By the property of the order, $k \mid \frac{p-1}{p_i}$ implies $g^{(p-1)/p_i} \equiv 1 \pmod{p}$. This contradicts our initial assumption that $g^{(p-1)/p_i} \not\equiv 1 \pmod{p}$ for all i .

Therefore, the assumption $k < p - 1$ is false, meaning $k = p - 1$. Hence, g is a primitive root. $\square$

Alg 12.1: Finding a Primitive Root. Let $p > 2$ be prime and $p - 1 = \prod_{i=1}^m p_i^{a_i}$ be the prime factorization of $p - 1$, with p_i distinct primes and $a_i \geq 1$ for $i = 1, \dots, m$.

1. Start with $g = 2$.
2. If $g^{(p-1)/p_i} \not\equiv 1 \pmod{p}$ for all $i = 1, \dots, m$, then g is a primitive root.
3. If not, increment $g \rightarrow g + 1$ and return to Step 2.

13 Number of Primitive Roots

Th 13.1: Characterization of Powers as Primitive Roots. Let p be a prime and g be a primitive root of $\mathbb{F}_p$. For any $i \in \{1, \dots, p-2\}$, we have:

1. g^i is a primitive root of $\mathbb{F}_p \iff \gcd(i, p-1) = 1$.
2. $\mathbb{F}_p$ has exactly $\phi(p-1)$ primitive roots.

Proof of 1. Let $p-1 = p_1^{a_1} \cdots p_m^{a_m}$ with p_j distinct primes and $a_j \geq 1$. By the effective test for a primitive root (Lemma from Lesson 5), g^i is not a primitive root $\iff (g^i)^{(p-1)/p_j} \equiv 1 \pmod{p}$ for some j . We analyze the condition $(g^i)^{(p-1)/p_j} \equiv 1 \pmod{p}$:

$$\begin{aligned} (g^i)^{(p-1)/p_j} \equiv 1 \pmod{p} &\iff \text{ord}_p(g) \mid \frac{i(p-1)}{p_j} \quad \text{since } g \text{ is a primitive root, } \text{ord}_p(g) = p-1 \\ &\iff p-1 \mid \frac{i(p-1)}{p_j} \\ &\iff p_j \mid i \end{aligned}$$

Thus, g^i is not a primitive root $\iff p_j \mid i$ for some j . Since the p_j are the distinct prime factors of $p-1$, the condition " $p_j \mid i$ for some j " is equivalent to $\gcd(i, p-1) > 1$. Therefore, g^i is a primitive root $\iff \gcd(i, p-1) = 1$. $\square$

Proof of 2. Since g is a primitive root, the elements of $\mathbb{F}_p^*$ are given by the distinct powers $\{1, g, \dots, g^{p-2}\}$. From part 1, the other primitive roots are the powers g^i such that $1 \leq i \leq p-2$ and $\gcd(i, p-1) = 1$. The number of such integers i in the range $\{1, \dots, p-2\}$ (or $\{1, \dots, p-1\}$) that are relatively prime to $p-1$ is exactly $\phi(p-1)$, where ϕ is Euler's totient function. Therefore, $\mathbb{F}_p$ has $\phi(p-1)$ primitive roots. $\square$

Lemma 13.1. Let $a, b, c \in \mathbb{Z}$ with $a \neq 0$. Then $a \mid bc \iff \frac{a}{\gcd(a,b)} \mid c$.

Proof. Let $d = \gcd(a, b)$. We set $a = a'd$ and $b = b'd$, where $\gcd(a', b') = 1$. The statement $a \mid bc$ is equivalent to $a'd \mid b'dc$. Since $d \neq 0$, we can cancel d : $a' \mid b'c$. Because $\gcd(a', b') = 1$, by Euclid's Lemma, $a' \mid b'c \iff a' \mid c$. Substituting back $a' = \frac{a}{d} = \frac{a}{\gcd(a,b)}$, we get the equivalence: $a \mid bc \iff \frac{a}{\gcd(a,b)} \mid c$. $\square$

Prop 13.1: The Order of a Power. Let p be a prime, g be a primitive root in $\mathbb{F}_p$, and $i \in \{1, \dots, p-2\}$. Then $\text{ord}_p(g^i) = \frac{p-1}{\gcd(p-1, i)}$.

Proof. The order $\text{ord}_p(g^i)$ is defined as $m = \min\{n \in \mathbb{N} : n \geq 1 \text{ so that } (g^i)^n \equiv 1 \pmod{p}\}$. The congruence $(g^i)^n \equiv 1 \pmod{p}$, or $g^{in} \equiv 1 \pmod{p}$, is equivalent to $\text{ord}_p(g) \mid in$. Since g is a primitive root, $\text{ord}_p(g) = p-1$. Thus, we are looking for:

$$\text{ord}_p(g^i) = \min\{n \in \mathbb{N} : n \geq 1 \text{ so that } (p-1) \mid in\}$$

Using the preceding Lemma with $a = p - 1$, $b = i$, and $c = n$, the condition $(p - 1) \mid in$ is equivalent to $\frac{p-1}{\gcd(p-1,i)} \mid n$. The minimum positive integer n such that $\frac{p-1}{\gcd(p-1,i)} \mid n$ is $n = \frac{p-1}{\gcd(p-1,i)}$. Therefore, $\text{ord}_p(g^i) = \frac{p-1}{\gcd(p-1,i)}$. $\square$

Cor 13.1: Alternative Proof. Let g be a primitive root in $\mathbb{F}_p$. Then g^i is a primitive root $\iff \gcd(p - 1, i) = 1$.

Proof. By the Characterization Theorem, g^i is a primitive root $\iff \text{ord}_p(g^i) = p - 1$. Using the Proposition on the order of a power:

$$\begin{aligned} \text{ord}_p(g^i) = p - 1 &\iff \frac{p - 1}{\gcd(p - 1, i)} = p - 1 \\ &\iff \gcd(p - 1, i) = 1 \end{aligned}$$

$\square$

Artin's Conjecture

Rem. If $g \in \mathbb{N}$ is a perfect square, there is no prime p such that $p > g$ and g is a primitive root in $\mathbb{F}_p$.

Th 13.2: Artin's Conjecture (Hooley, 1967 - under GRH). If $g \geq 2$ is not a perfect square, the set of primes p for which g is a primitive root in $\mathbb{F}_p$ has a positive density.

$$\lim_{x \rightarrow +\infty} \frac{\#\{p \leq x : p \text{ is prime and } g \text{ is a primitive root in } \mathbb{F}_p\}}{\#\{p : p \text{ is prime, } p \leq x\}} = C_g > 0$$

For $g = 2$, the constant is $C_2 \approx 0.3739558 \dots$

Th 13.3: Roger-Heath-Brown, 1985 - unconditional. There are at most two primes that are not primitive roots for infinitely many primes.

Rem. This theorem implies that among the first few primes, like 2, 3, and 5, at least one of them must be a primitive root for infinitely many primes.